CONORMAL ALGEBRAS OF THE FIBRES INDUCED BY A FORMAL LOCAL MAP ARE CONTROLLED BY ONE FINITE TYPE MODIFICATION

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Editorial note. Statement span is the whole lemma environment lines 11570-11590 including the displayed pair of closed immersions; proof span is lines 11592-11718.

Editorial note. The proof cites several labelled results, which are prerequisites and are supplied as non-load-bearing navigation stubs.

Editorial note. The author's prose contains an informal interjection; the text is retained verbatim.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Noetherian schemes of finite type, completions of local rings at their maximal ideal, base change of a morphism along a formal local map, and conormal algebras of closed immersions cut out by such base changes. The control construction and the comparison of conormal algebras are retained.

Source reference. Original lemma lemma-control-agreement: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11530–11542. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07VF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11718–11718. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07VD: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11718–11718. Referenced by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $Y \to T \to S$ be finite type morphisms of Noetherian schemes. Let $t \in T$ map to $s \in S$ and let $\sigma : \mathcal{O}_{T,t} \to \mathcal{O}^\wedge_{S,s}$ be a local $\mathcal{O}_{S,s}$-algebra map. There exists a finite type morphism $(T', t') \to (T, t)$ such that σ factors through $\mathcal{O}_{T,t} \to \mathcal{O}_{T',t'}$ and such that for every local $\mathcal{O}_{S,s}$-algebra map $\sigma' : \mathcal{O}_{T,t} \to \mathcal{O}^\wedge_{S,s}$ which factors through $\mathcal{O}_{T,t} \to \mathcal{O}_{T',t'}$ the closed immersions*

$$Y \times_{T,\sigma} \operatorname{Spec}(\mathcal{O}^\wedge_{S,s}) = Y_\sigma \longleftarrow Y_t \longrightarrow Y_{\sigma'} = Y \times_{T,\sigma'} \operatorname{Spec}(\mathcal{O}^\wedge_{S,s})$$

have isomorphic conormal algebras.

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Proof. A useful observation is that $\kappa(s) = \kappa(t)$ by the existence of σ. Observe that the statement makes sense as the fibres of Y_σ and $Y_{\sigma'}$ over $s \in \mathrm{Spec}(\mathcal{O}^\wedge_{S,s})$ are both canonically isomorphic to Y_t. We will think of the property "σ' factors through $\mathcal{O}_{T,t} \to \mathcal{O}_{T',t'}$" as a constraint on σ'. If we have several such constraints, say given by $(T'_i, t'_i) \to (T, t)$, $i = 1, \ldots, n$ then we can combined them by considering $(T'_1 \times_T \ldots \times_T T'_n, (t'_1, \ldots, t'_n)) \to (T, t)$. We will use this without further mention in the following.

By Lemma lemma-control-agreement we can assume that any σ' as in the statement of the lemma is the same as σ modulo $\mathfrak{m}_s^2$. Note that the conormal algebra of Y_t in Y_σ is just the quasi-coherent graded $\mathcal{O}_{Y_t}$-algebra

$$\bigoplus\nolimits_{n\geq 0} \mathfrak{m}_s^n \mathcal{O}_{Y_\sigma}/\mathfrak{m}_s^{n+1}\mathcal{O}_{Y_\sigma}$$

and similarly for $Y_{\sigma'}$. Since σ and σ' agree modulo $\mathfrak{m}_s^2$ we see that these two algebras are the same in degrees 0 and 1. On the other hand, these conormal algebras are generated in degree 1 over degree 0. Hence if there is an isomorphism extending the isomorphism just constructed in degrees 0 and 1, then it is unique.

We may assume S and T are affine. Let $Y = Y_1 \cup \ldots \cup Y_n$ be an affine open covering. If we can construct $(T'_i, t'_i) \to (T, t)$ as in the lemma such that the desired isomorphism (see previous paragraph) exists for $Y_i \to T \to S$ and σ, then these glue by uniqueness to prove the result for $Y \to T$. Thus we may assume Y is affine.

Write $S = \mathrm{Spec}(R)$, $T = \mathrm{Spec}(C)$, and $Y = \mathrm{Spec}(B)$. Choose a presentation $B = C[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$. Denote $R^\wedge = \mathcal{O}^\wedge_{S,s}$. Let $a_{kj} \in R^\wedge[x_1, \ldots, x_n]$ be polynomials such that

$$\sum\nolimits_{j=1,\ldots,m} a_{kj}\sigma(f_j) = 0, \quad \text{for } k = 1, \ldots, K$$

is a set of generators for the module of relations among the $\sigma(f_j) \in R^\wedge[x_1, \ldots, x_n]$. Thus we have an exact sequence

(0.1.1)

$$R^\wedge[x_1, \ldots, x_n]^{\oplus K} \to R^\wedge[x_1, \ldots, x_n]^{\oplus m} \to R^\wedge[x_1, \ldots, x_n] \to B \otimes_{C,\sigma} R^\wedge \to 0$$

Let c be an integer which works in the Artin-Rees lemma for both the first and the second map in this sequence and the ideal $\mathfrak{m}_{R^\wedge} R^\wedge[x_1, \ldots, x_n]$ as defined in More on Algebra, Section 07VD. Write

$$a_{kj} = \sum\nolimits_{I\in\Omega} a_{kj,I} x^I \quad \text{and} \quad f_j = \sum\nolimits_{I\in\Omega} f_{j,I} x^I$$

in multiindex notation where $a_{kj,I} \in R^\wedge$, $f_{j,I} \in C$, and Ω a finite set of multiindices. Then we see that

$$\sum\nolimits_{j=1,\ldots,m,\ I,I'\in\Omega,\ I+I'=I''} a_{kj,I}\sigma(f_{j,I'}) = 0, \quad I'' \text{ a multiindex}$$

in $R^\wedge$. Thus we take

$$C' = C[t_{jk,I}]/\left(\sum\nolimits_{j=1,\ldots,m,\ I,I'\in\Omega,\ I+I'=I''} t_{kj,I} f_{j,I'},\ I'' \text{ a multiindex}\right)$$

Then σ factors through a map $\tilde{\sigma} : C' \to R^\wedge$ sending $t_{kj,I}$ to $a_{jk,I}$. Thus $T' = \mathrm{Spec}(C')$ comes with a point $t' \in T'$ such that σ factors through $\mathcal{O}_{T,t} \to \mathcal{O}_{T',t'}$. Let $t_{kj} = \sum t_{kj,I} x^I$ in $C'[x_1, \ldots, x_n]$. Then we see that we have a complex

$$C'[x_1, \ldots, x_n]^{\oplus K} \to C'[x_1, \ldots, x_n]^{\oplus m} \to C'[x_1, \ldots, x_n] \to B \otimes_C C' \to 0 \tag{0.1.2}$$

which is exact at $C'[x_1, \ldots, x_n]$ and whose base change by $\tilde{\sigma}$ gives (0.1.1).

By Lemma lemma-control-agreement we can find a further morphism $(T'', t'') \to (T', t')$ such that $\tilde{\sigma}$ factors through $\mathcal{O}_{T',t'} \to \mathcal{O}_{T'',t''}$ and such that if $\sigma' : C \to R^\wedge$

factors through $\mathcal{O}_{T'',t''}$, then the induced map $\tilde{\sigma}' : C' \to R^\wedge$ agrees modulo $\mathfrak{m}_s^{c+1}$ with $\tilde{\sigma}$. Thus if σ' is such a map, then we obtain a complex

$$R^\wedge[x_1, \ldots, x_n]^{\oplus K} \to R^\wedge[x_1, \ldots, x_n]^{\oplus m} \to R^\wedge[x_1, \ldots, x_n] \to B \otimes_{C,\sigma'} R^\wedge \to 0$$

over $R^\wedge[x_1, \ldots, x_n]$ by applying $\tilde{\sigma}'$ to the polynomials t_{kj} and f_j. In other words, this is the base change of the complex (0.1.2) by $\tilde{\sigma}'$. The matrices defining this complex are congruent modulo $\mathfrak{m}_s^{c+1}$ to the matrices defining the complex (0.1.1) because $\tilde{\sigma}$ and $\tilde{\sigma}'$ are congruent modulo $\mathfrak{m}_s^{c+1}$. Since (0.1.1) is exact, we can apply More on Algebra, Lemma 07VF to conclude that

$$\mathrm{Gr}_{\mathfrak{m}_s}(B \otimes_{C,\sigma'} R^\wedge) \cong \mathrm{Gr}_{\mathfrak{m}_s}(B \otimes_{C,\sigma} R^\wedge)$$

as desired. □